Mohamed Zakaria Elnemr

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SUMMARY

Final-year Computer Science student specializing in Machine Learning and Generative AI, with hands-on experience developing end-to-end ML pipelines, computer vision applications, and RAG-powered systems. Experienced in building practical AI solutions using Python, PyTorch, Scikit-learn, and modern LLM frameworks, with training experience through DEPI, ITI, ITIDA/NTI, and Edrak AI.

EDUCATION

Mansoura University | *B.Sc. in Computer and Information Sciences* Sep 2023 – 2027 (Expected)

- **Final Year (4th Year)** | **Cumulative Grade: Very Good (83.0%)**
- **Relevant Coursework:** Artificial Intelligence, Algorithm Analysis & Design, Software Engineering, Network & Computer Security.

EXPERIENCE

AI and Machine Learning Trainee | *Information Technology Institute (ITI)* Aug 2026 – Sep 2026

- **Completed** an intensive 90-hour certified training program on AI and Machine Learning Foundations covering Python, NumPy, Probability, Linear Algebra, Data Exploration, Machine Learning, and Deep Learning.
- **Delivered** a successful Final Project & Demo applying core machine learning and neural network workflows.

Machine Learning Trainee | *ITIDA & NTI* Aug 2026 – Sep 2026

- **Completed** a comprehensive summer training program (120 total hours: 90 technical, 30 freelancing) focusing on applied machine learning algorithms and workflows.
- **Achieved** an overall training score of 95.5%.

Generative AI Intern | *Edrak For AI* May 2026 – Jul 2026

- **Completed** a comprehensive 126-hour Generative AI internship focused on modern AI concepts, LLM applications, and practical Generative AI workflows.
- **Built** end-to-end AI applications leveraging LLMs, prompt engineering, and retrieval-augmented generation (RAG) to solve practical business scenarios.

Machine Learning Engineer Trainee (Microsoft Track) | *DEPI* Jun 2025 – Dec 2025

- **Engineered** and deployed end-to-end ML pipelines on Azure, applying MLOps practices for model deployment, experiment tracking, versioning, and monitoring using MLflow.
- **Developed** Deep Learning models for NLP and Computer Vision tasks, applying transfer learning and attention architectures to improve inference accuracy.
- **Built** Generative AI solutions and RAG-based applications utilizing advanced Prompt Engineering techniques tailored for diverse datasets.

SELECTED PROJECTS

MUSTAWFI – Smart Loan Eligibility Platform | *Python, FastAPI, OpenCV, Scikit-learn, React, TanStack Start* GitHub

- **Developed** an end-to-end loan application processing system using OpenCV-based document alignment and custom OCR to extract structured data from handwritten and printed forms.
- **Integrated** Human-in-the-Loop verification with Scikit-learn classification models to predict loan eligibility, generate confidence scores, and support bilingual Arabic/English workflows.

Bayan AI | *RAG, Qwen 2.5, FastAPI, React, TypeScript* GitHub

- **Built** an AI-powered cross-cultural communication assistant that rewrites professional messages according to the target culture's etiquette while preserving the original meaning.
- **Implemented** a RAG pipeline utilizing FAISS to retrieve relevant cultural knowledge, construct contextual prompts, and generate tailored responses through a FastAPI backend.

NosoosBot | *Python, Aiogram, Groq, Llama 3.3 70B* GitHub

- **Developed** an asynchronous Telegram bot for extracting YouTube captions, translating transcripts, and generating AI-powered multilingual summaries.
- **Implemented** transcript caching, rate limiting, document exports (PDF/DOCX), and Map-Reduce summarization techniques to process long-form content efficiently.

Hand Gesture Recognition System | *Python, OpenCV, MediaPipe* GitHub

- **Engineered** a real-time hand gesture recognition system using Python and MediaPipe to track and classify predefined gestures.
- **Implemented** computer vision pipelines using OpenCV to process hand landmarks and enable real-time gesture inference.

BHAAV – House Price Prediction | *Scikit-learn, TypeScript, React* GitHub

- **Developed** an AI-powered platform for predicting house prices across the Indian real-estate market using machine learning regression algorithms.
- **Architected** the end-to-end ML workflow, from data preprocessing and model training to evaluation and API integration into a React-based web application.

TECHNICAL SKILLS

Programming: Python, SQL, C++, PHP

ML/DL: Scikit-learn, PyTorch, TensorFlow, NumPy

GenAI/NLP: RAG, LLMs, Hugging Face, FAISS, Prompt Engineering

Computer Vision: OpenCV, MediaPipe, OCR

Backend & Frontend: FastAPI, Django, React, TypeScript

Cloud & MLOps: Azure AI, MLflow, Docker, Git, Linux

CERTIFICATIONS

Fundamentals of Deep Learning — NVIDIA *Aug 2026*

Computer Vision — Kaggle Learn *Aug 2026*

Intro to Deep Learning — Kaggle Learn *Aug 2026*

Generative AI (Beginner Level) — NVIDIA DLI & ITI *Sep 2025*

HCIA AI Learning Course (40 Hours) — Huawei ICT Academy *Summer 2025*